

IoT Platform Programming and Design for Smart Home Applications

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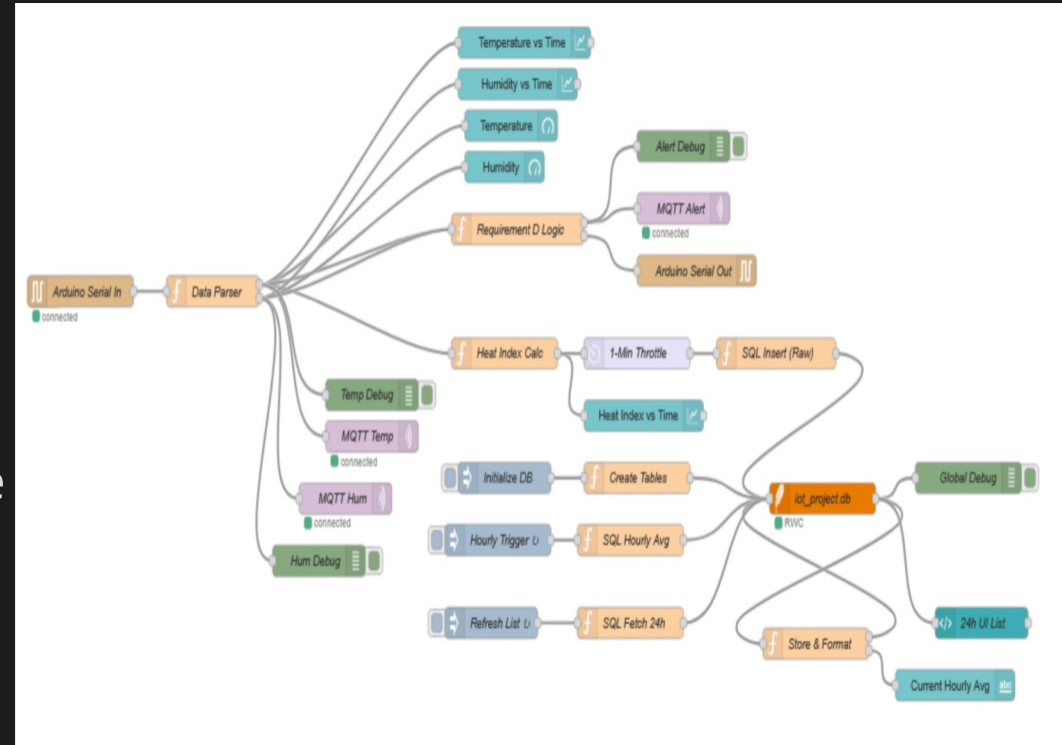
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Project Objectives & Requirements

- **Requirement A:** Real-time visualization of Temperature, Humidity and Heat Index.
- **Requirement B:** Persistent data logging at 1-minute intervals using SQLite.
- **Requirement C:** Hourly statistical aggregation (Averages) and historical logging for 24 hours.
- **Requirement D:** Intelligent control loop and alert system via Serial and MQTT.

System Architecture

- **Modular 3-Layer Approach:**
 - **Perception Layer:** Arduino Uno + DHT22 sensor for telemetry.
 - **Processing Layer:** Node-RED orchestrator for data parsing and logic.
 - **Application Layer:** User Dashboard and SQLite database for analytics.
- **Hybrid Communication:** Serial (UART) for low-latency local feedback and MQTT for remote notifications.



Core Technologies & Communication Stack

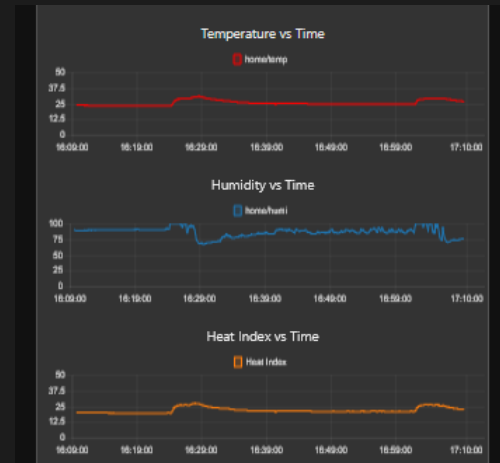
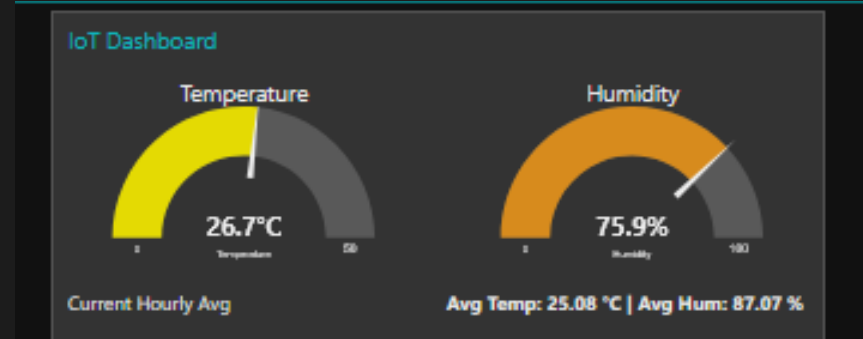
- **Edge Computing**: Arduino Uno with a DHT22 sensor for high-precision environmental telemetry.
- **Orchestration**: Node-RED environment installed locally, acting as the central processing hub for data parsing and logic.
- **Local MQTT Broker**: Eclipse Mosquitto installed on the host machine.
 - ▶ Ensures data privacy and system autonomy.
 - ▶ Facilitates asynchronous messaging for the alert system.

Core Technologies & Communication Stack

- **Storage Engine**: SQLite 3 for lightweight, serverless data persistence and historical analytics.
- **Protocol Synergy**:
 - ▶ **Serial (UART)**: Dedicated link for near-zero latency physical feedback (LED control).
 - ▶ **MQTT**: Scalable protocol for network-based notifications and future multi-device expansion.

Real-Time Monitoring

- **Node-RED Dashboard:**
Utilizing radial gauges for instant readings and line charts for temporal trends.
- **Feature Engineering:**
Calculation of the *Heat Index (HI)* within Node-RED to assess human comfort levels.



Data Persistence & Analytics

- **Data Logging:** Telemetry is stored in the *sensor_data* table every 1 minute.
- **Automated Aggregation:** An hourly trigger calculates the mean values using SQL `AVG()`.
- **Historical View:** Results are archived in *hourly_stats* and displayed in a structured UI list (Last 24h)

Database Structure Browse Data Edit Pragma Execute SQL

Table: sensor_data

	id	timestamp	temperature	humidity	heat_index
	Filter	Filter	Filter	Filter	Filter
37	37	2026-02-13 13:50:38	25.1	90.8	21.58
38	38	2026-02-13 13:51:38	24.9	86.6	21.16
39	39	2026-02-13 13:52:38	24.8	86.2	21.03
40	40	2026-02-13 13:53:38	24.7	87.7	20.99
41	41	2026-02-13 13:54:38	24.6	87.3	20.86
42	42	2026-02-13 13:55:38	24.5	87.3	20.75
43	43	2026-02-13 13:56:38	24.4	88.3	20.69
44	44	2026-02-13 13:57:38	24.3	88.2	20.58
45	45	2026-02-13 13:58:39	24.3	88.2	20.58
46	46	2026-02-13 13:59:39	24.3	88.2	20.58
47	47	2026-02-13 14:00:39	24.1	88.1	20.35

Database Structure Browse Data Edit Pragma Execute SQL				
Table: hourly_stats				
	id	timestamp	avg_temp	avg_hum
	Filter	Filter	Filter	Filter
1	1	2026-02-13 13:23:15	25.79	92.56
2	2	2026-02-13 13:25:21	26.02	90.22
3	3	2026-02-13 13:25:24	26.02	90.22
4	4	2026-02-13 13:32:17	26.04	91.99
5	5	2026-02-13 13:48:39	25.54	91.5
6	6	2026-02-13 14:50:31	25.08	87.07

Hourly Average History (Last 24h)				
2026-02-13 14:50:31	→	25.08°C		
2026-02-13 13:48:39	→	25.54°C		
2026-02-13 13:32:17	→	26.04°C		
2026-02-13 13:25:24	→	26.02°C		
2026-02-13 13:25:21	→	26.02°C		
2026-02-13 13:23:15	→	25.79°C		

Automation Logic & Safety

- **Threshold Condition:** Alert triggers if *Temperature* > 30°C and *Humidity* > 70%.
- **Dual Action Response:**
 - ▶ **Local:** *LED_ON* command sent via Serial to the Arduino.
 - ▶ **Network:** *alert* payload published to MQTT topic *home/alert*.
- **Automatic Reset:** System sends *stopalert* / *LED_OFF* when conditions normalize.

```
SELECT timestamp, avg_temp, avg_hum FROM hourly_stats ORDER BY timestamp
DESC LIMIT 24; : msg.payload : array[6]

▶ [ object, object, object, object, object, object ]

2/13/2026, 5:06:14 PM  node: Alert Debug
home/alert : msg.payload : string[5]

"alert"

2/13/2026, 5:06:14 PM  node: Alert Debug
home/alert : msg.payload : string[5]

"alert"

2/13/2026, 5:06:16 PM  node: Global Debug
INSERT INTO sensor_data (temperature, humidity, heat_index, timestamp) VALUES (29.1,
72.8, 25.1... : msg.payload : array[0]

[ empty ]

2/13/2026, 5:06:19 PM  node: Alert Debug
home/alert : msg.payload : string[9]

"stopalert"

2/13/2026, 5:06:20 PM  node: Alert Debug
home/alert : msg.payload : string[9]

"stopalert"
```


Conclusions & Key Outcomes

- **Successful integration of hardware sensing with database persistence and cloud-like middleware.**
- **Reliable automation with minimal latency due to hybrid communication model.**
- **All project requirements (A, B, C, and D) fully implemented and validated.**

Thank you

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